

Weekly meeting progress

Friday, Jul 24, 2026





What is the best solution between stopping the episode when an agent exceeds the limit or giving it a penalty?

What is the best solution between the following:

- Stop the episode when an agent exceeds the limit
- Give the agent a penalty when it exceeds the limit



What is the best solution between stopping the episode when an agent exceeds the limit or giving it a penalty?

What is the best solution between the following:

- Stop the episode when an agent exceeds the limit
 - This weakens the router's learning because the agent reaches the limit approximately every 5 steps.
- Give the agent a penalty when it exceeds the limit
 - The agent has difficulty getting rid of indexes selected at the beginning of the episode.



Another solution proposed by Sam

Train the agent without any budget limit, and at the end propose a set of indexes that fits within the budget.

But how would the router know the real execution time?



Another solution proposed by Sam

Train the agent without a hard storage limit, and at the end of each episode, propose a subset of indexes that fits within the budget (using a knapsack selection).

But how would the router know the real execution time?

- The worker computes the cost using only the selected indexes (the best ones chosen based on the agent's gains) and sends this cost to the router. This way, the router learns from realistic execution times.



First benchmark results with no space budget during the training

- With 4 replicates
- Total runtime: 3 304.846s (DINA and qDINA: 3000s)
- With 38 indexes:
 - Replica 0: 8
 - Replica 1: 16
 - Replica 2: 10
 - Replica 3: 4



Limitations of this solution

It requires estimating execution time twice:

- Once to compute the agent's reward using all selected indexes
- Once to evaluate the actual execution time using only the subset of indexes that fits within the storage budget



Next tasks

- Continue improving the penalty-based solution
- Explore ways to enhance the new approach